

Newton Laws of motion.

Inertia — that property of matter which causes it to resist change in its state of motion.

Force — is a push or pull exerted on an object in any direction. That is, force has direction and magnitude. $\rightarrow$ vector.

Newton's Laws.

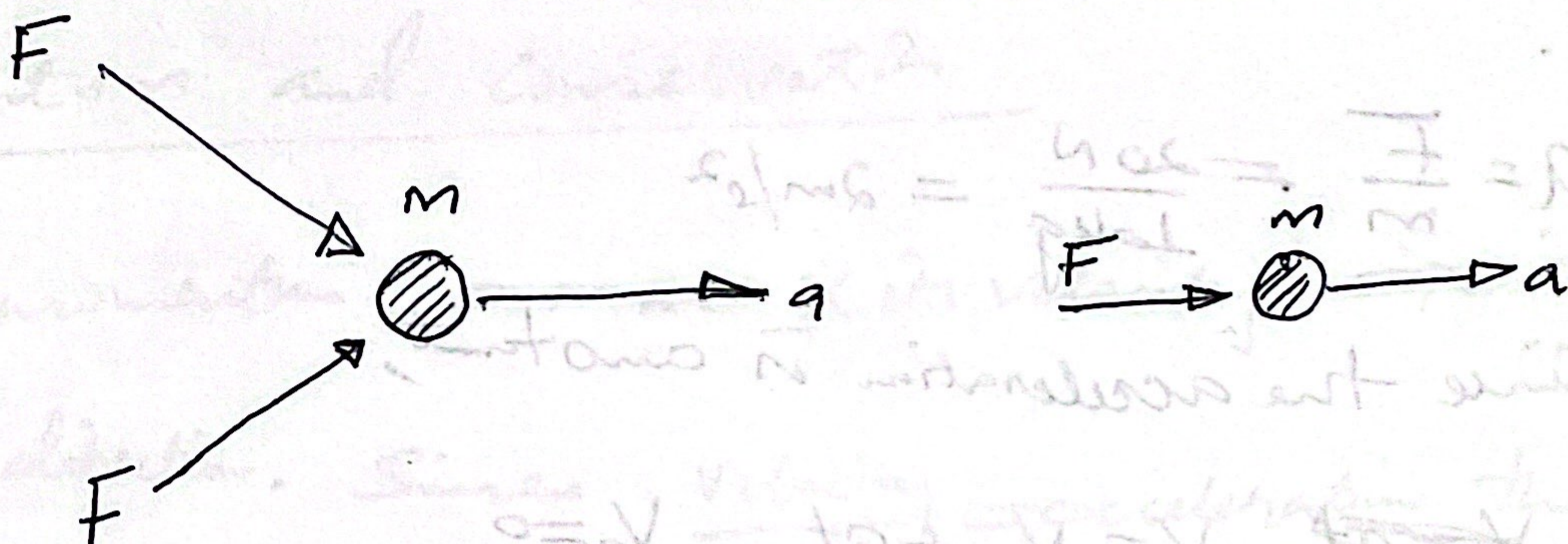
I. Law of inertia — If the net force on an object is zero then the acceleration of the object is zero and the object moves with constant velocity.

i.e. if $F=0$ then $a=0$ or $v=\text{constant}$

This law defines natural motion for isolated bodies. It is valid even in the domain of ^{Quantum} ~~classical~~ mechanics.

II. Law of Force & acceleration — The accelerated motion of a body ^{is proportional to the applied} ~~can only be produced by a force.~~ And the direction of acceleration is along the direction of the force.

$$F = ma$$



Case of net Force

III Law of reaction — If object 1 exerts a force on object 2, then object 2 exerts an equal force oppositely directed, on object 1.

i.e. $\vec{F}_{12} = -\vec{F}_{21}$

Units of force $\rightarrow N = 1 \text{ kgm/s}^2$. It is the force required to give a mass of 1 kg an acceleration of 1 m/s^2 .

~~* Noted proportion~~

Example:

An object of mass 10 kg is at rest. A net force of 20 N is applied for 1 s . What is the final velocity? How far will the object have moved in the 1 s interval?

Sol.

$$a = \frac{F}{m} = \frac{20 \text{ N}}{10 \text{ kg}} = 2 \text{ m/s}^2$$

Since the acceleration is constant,

$$\cancel{V=at} \quad V = V_0 + at, \quad V_0 = 0$$

$$\Rightarrow V = at = (2 \text{ m/s}^2)(10 \text{ s}) = 20 \text{ m/s}$$

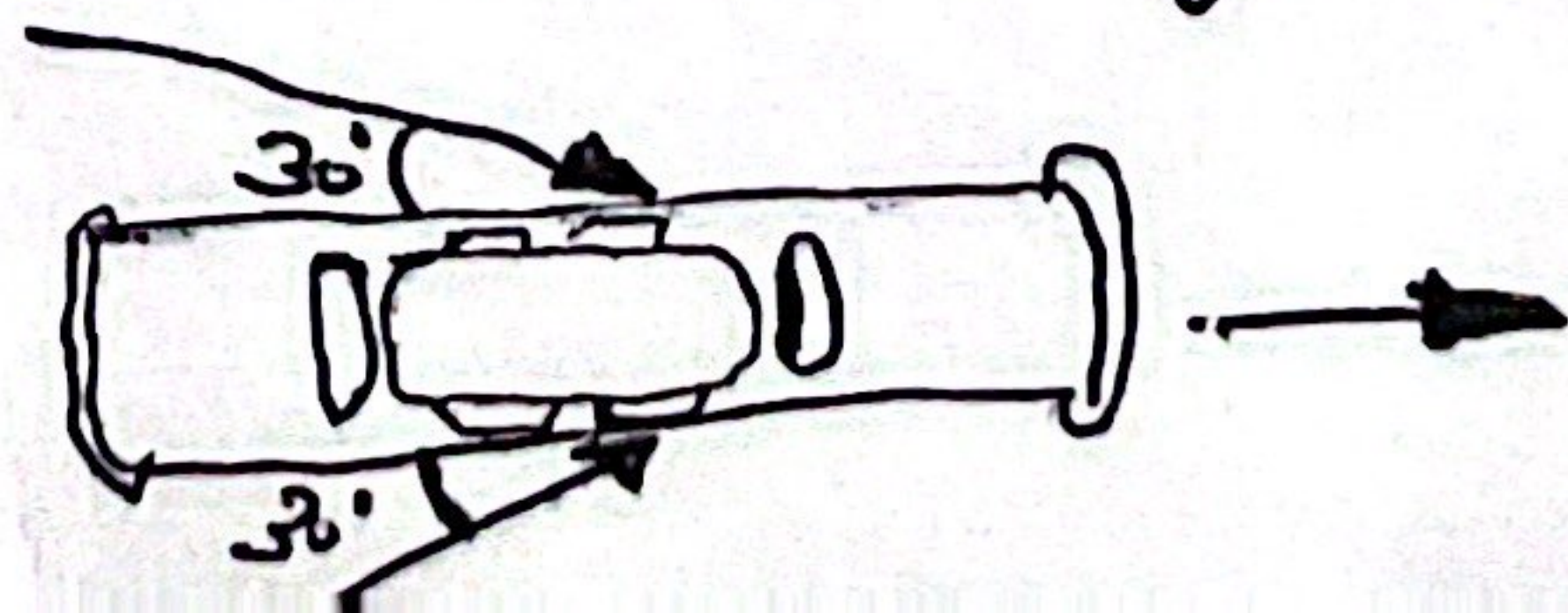
$$\text{distance } X = V_0 t + \frac{1}{2} at^2 = \frac{1}{2} (2 \text{ m/s}^2)(10 \text{ s})^2 = 100 \text{ m}$$

Exercise.

□ Shortly after jumping from an airplane, a skydiver has a downward acceleration of 9 m/s^2 . Later his downward acceleration is 1 m/s^2 . Has the net force acting on him changed? If so, by what factor?

□ A jet plane starts its take off roll with an acceleration of 2 m/s^2 . Each of its two jet engines delivers a thrust of 10^5 N . What is the approximate mass of the plane.

□ The mass of a stalled car is 1300 kg . 2 men push on the car as shown. If their efforts results in an acceleration of the car of 0.03 m/sec^2 What is the force ~~acting~~ applied by each man if both apply equal force.



Momentum and Conservation

Linear momentum $P = \text{mass} \times \text{its velocity} \Rightarrow \text{units kg m/s}$

It has a direction. Since $\frac{\text{Velocity}}{\text{time}} = \text{acceleration}$ then

~~$P = M \cdot V$~~

$$\text{Force } F = \text{Mass} \times \frac{\text{Velocity}}{\text{time}} = \frac{P}{t}$$

$$\therefore \boxed{\bar{F} = \frac{\Delta P}{\Delta t}}$$

Which gives another way of defining Newton's II law. [Force is time rate of change of momentum].

Example

A car of mass 1600 kg travelling at a speed of 20 m/s comes to rest in 5 s. What average force did the ^{brakes} ~~brakes~~ exert on the car? What was its original P?

Sol.

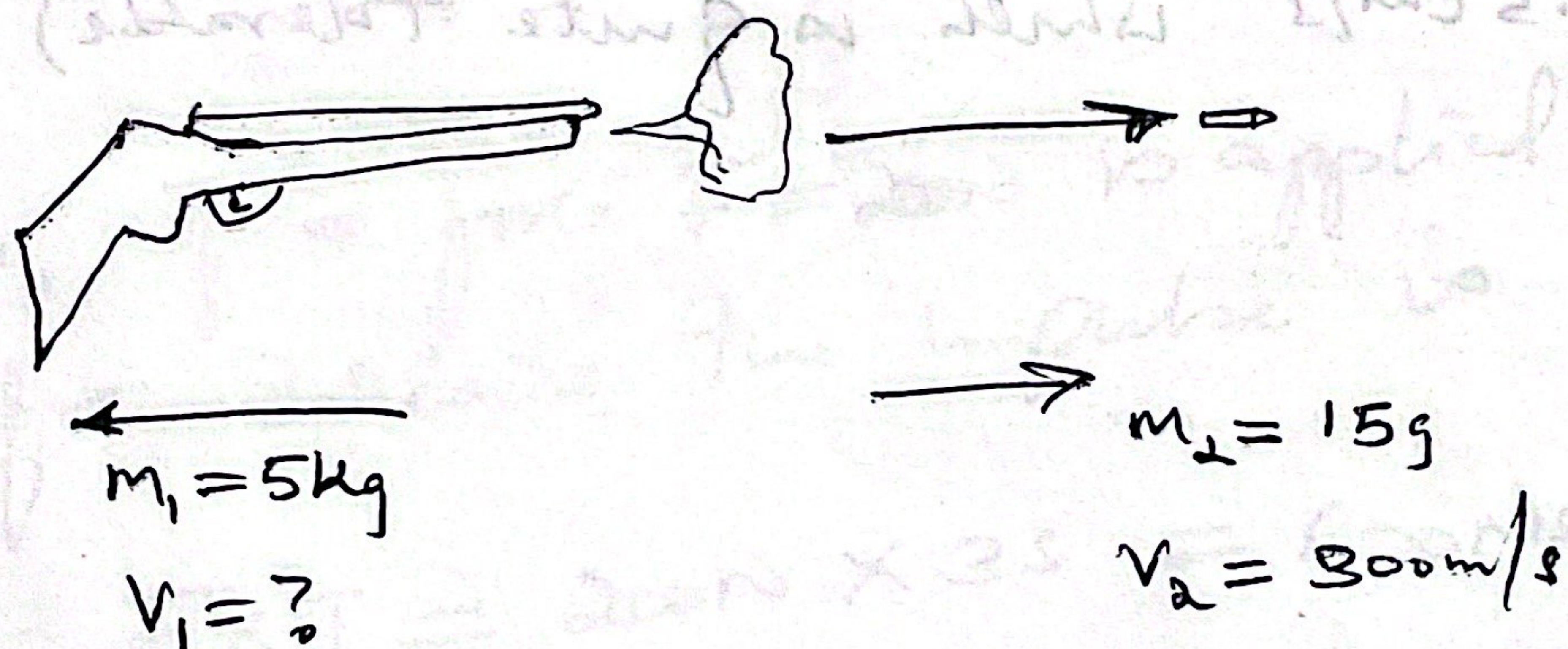
$$\text{Momentum } P = M V = 1600 \text{ kg} \times 20 \text{ m/s} = 32000 \text{ kg m/s}$$

$$\text{Force} = \frac{\Delta P}{\Delta t} = \frac{32000 \text{ kg m/s}}{5 \text{ s}} = \underline{\underline{6400 \text{ N}}}$$

Example

A high powered rifle whose mass is 5kg fires a 15g bullet with a muzzle velocity of 300m/s. What is the Recoil Velocity of the rifle.

Sol.



From the conservation law we have

$$M_1 V_1 = -m_2 V_2$$

$$V_1 = -\frac{m_2 V_2}{m_1}$$

$$= -\frac{(0.015\text{kg}) \times (300\text{m/s})}{5\text{kg}}$$

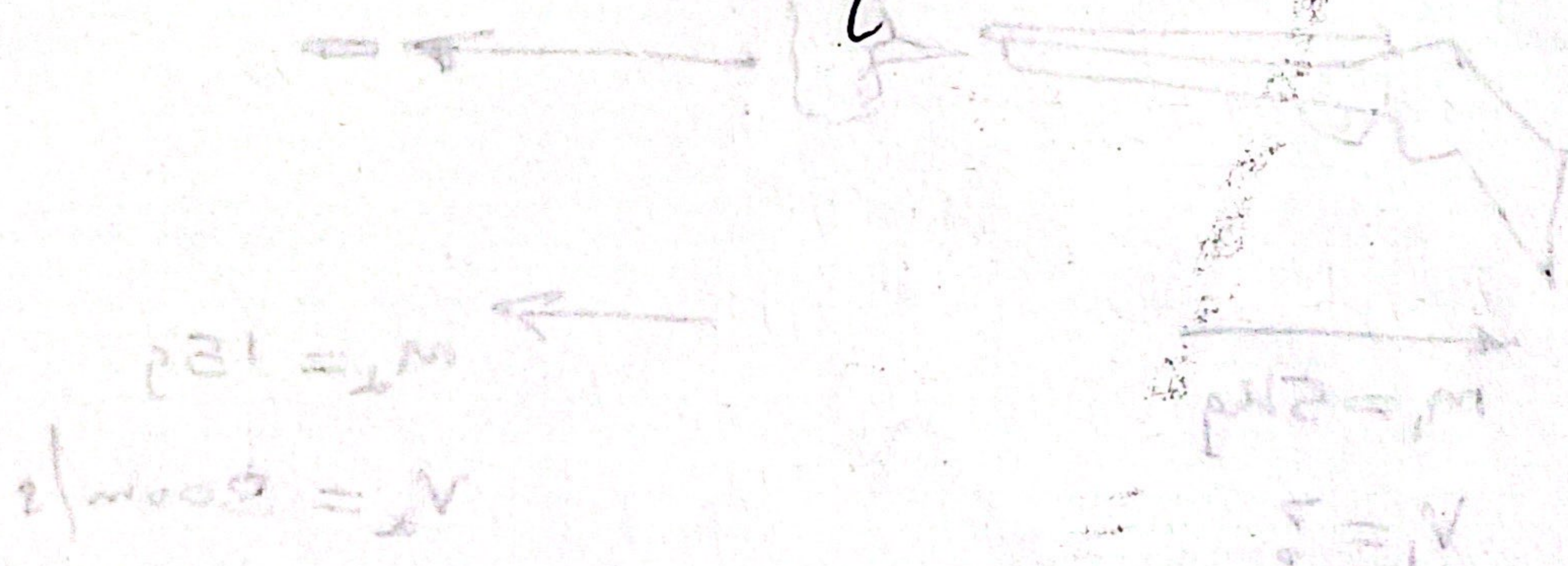
$$\underline{\underline{V_1 = -0.9\text{m/s}}} \quad \text{Which is substantial.}$$

* Rocket propulsion utilizes momentum conservation. To accelerate the payload, mass must be thrown away.

Exercise.

In the above example, Suppose the shooter holds firmly holds the rifle ~~firmly~~ against his shoulder, the shooters body as a whole absorbs the momentum. If now we use M_1 as the mass of the rifle plus the mass of the shooter (Looky), ~~find~~ what is the recoil velocity?

($V_1 \approx -4.5 \text{ cm/s}$ which is quite tolerable)



From the conservation law we have

$$V_1 = -\frac{m_b v_b}{m}$$

$$V_1 = -\frac{(0.012 \text{ kg}) \times (300 \text{ m/s})}{2 \text{ kg}}$$

$$V_1 = -0.18 \text{ m/s} \quad \text{which is tolerable}$$

Correct proportion of forces must be taken into account. The recoil velocity is not the same as the bullet velocity.

Note

Impulse

Impulse is defined by $I = \int_{t_1}^{t_2} F dt$ or simply

$$I = \text{Force}(F) \times \text{time}(t)$$

It is when the changes in time ^{are small} that the integral is used.

For example, if a force $F = 200\text{N}$ is applied to a car by the engine for 3 seconds the impulse is

$$I = F \times T = 200\text{N} \times 3\text{s} = 600\text{Ns}$$

Note If we are talking about a system of several particles moving together, we talk about the Centre of mass of the system (which as we saw coincides with CG) as moving with a velocity V . The Centre of mass of a system behaves in every way like a particle of mass M experiencing force F_{ext} . Therefore Newton's I and II law should be restated in terms of the centre of mass, & if momentum is conserved, the Centre of mass velocity is constant.

Inelastic and Elastic Collisions.

- In collisions, the total momentum of the colliding objects is always conserved. Usually, however, their total kinetic energy is not conserved, so $\rightarrow$ it is changed to heat or sound energy, which is not recoverable. Such collision is said to be inelastic.

Inelastic collision = collision where total k.e is not conserved (total momentum always conserved in any type of collision)

If the total kinetic energy is conserved, the collision is said to be elastic. Gas molecules, such as the air molecules in a room, make elastic collisions. The collision between two snooker balls is approx elastic.

Elastic collision = collision where total k.e is conserved

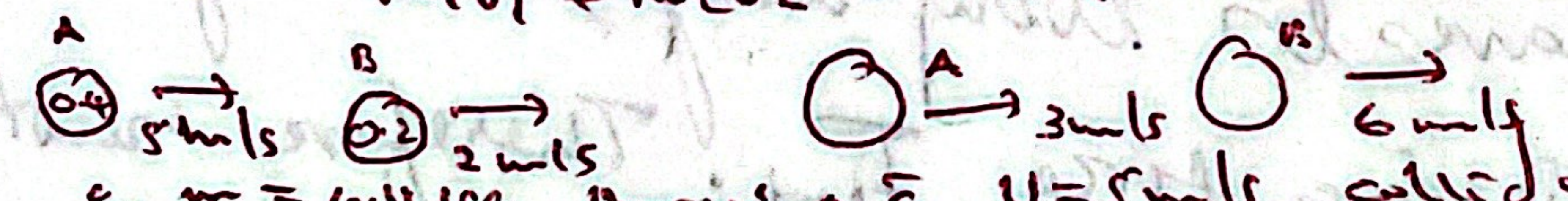
- 2 masses m_1 and m_2 , moving $\rightarrow$ diff. velocities u_1 and u_2 collide head-on with $\odot$ over.

- After collision their velocity $\rightarrow$ v_1 and v_2 respectively. If collision is an elastic one, their total k.e is conserved. So

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

also for conservation of momentum which applies to any type of collision

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

eg. 
A ball A $m = 0.4 \text{ kg}$ moving $\rightarrow$ $u = 5 \text{ m/s}$ collides head-on with ball B mass 0.2 kg moving $\leftarrow$ $u = 2 \text{ m/s}$. After collision A moves at 3 m/s and B $\leftarrow$ $v = 6 \text{ m/s}$.
(i) show the collision obeys law of conservation of momentum
(ii) is collision elastic/inelastic.

Be4 $(0.4 \times 5) + (0.2 \times 2) = 2.4 \text{ Ns}$. After $(0.4 \times 3) + (0.2 \times 6) = 2.4 \text{ Ns}$
 $\therefore$ momentum is conserved.

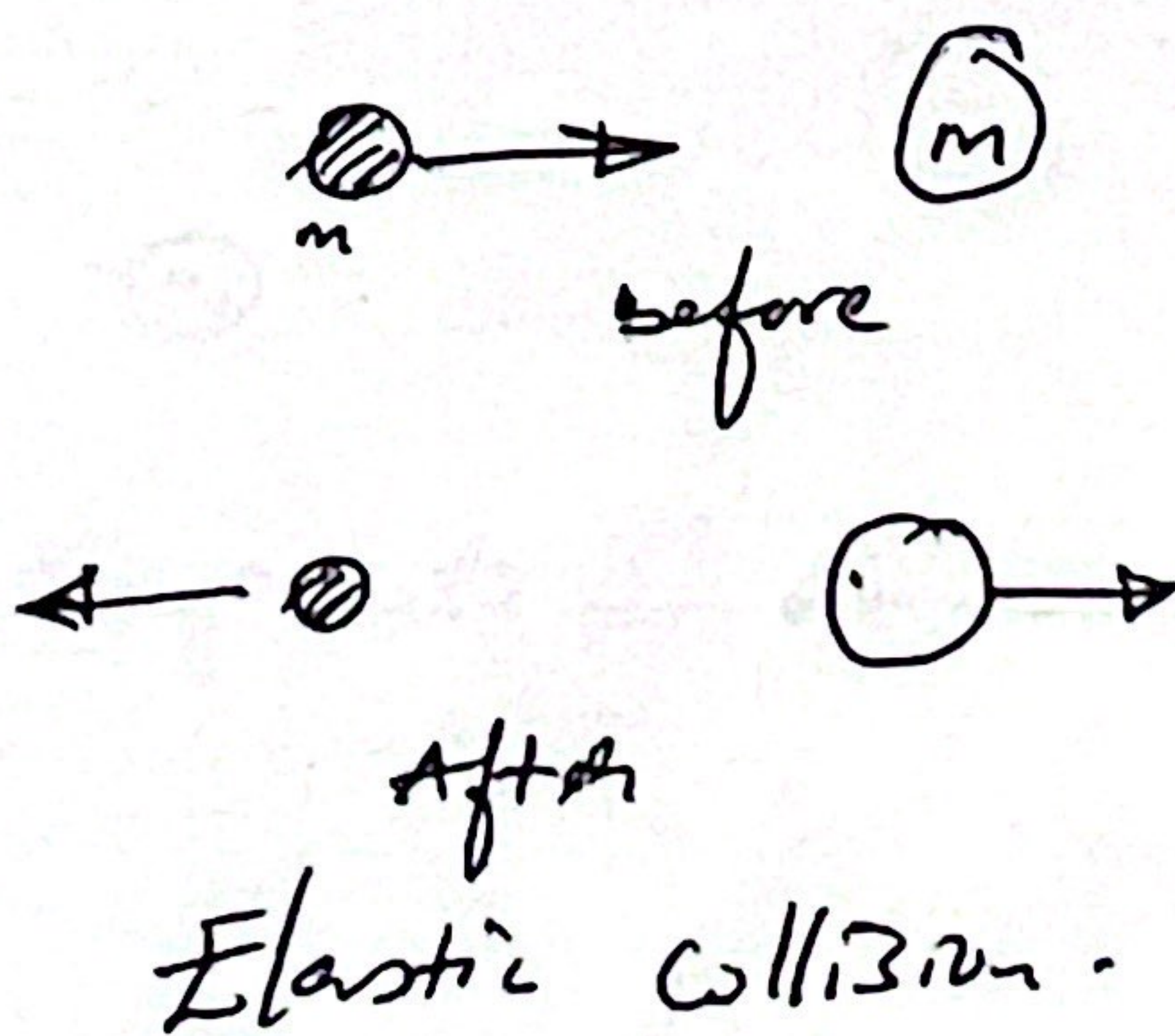
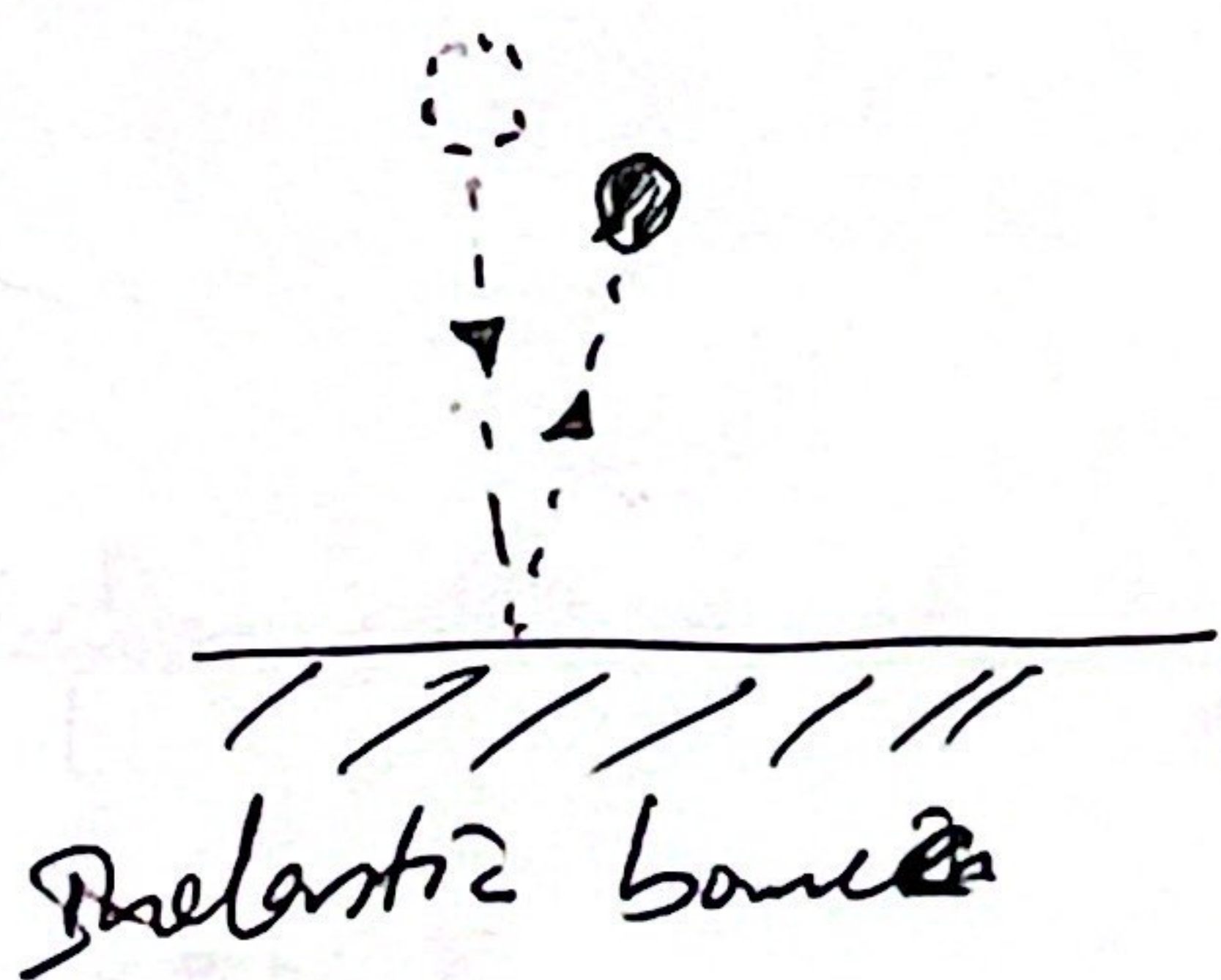
(iii) Be4 $(\frac{1}{2} \times 0.4 \times 5^2) + (\frac{1}{2} \times 0.2 \times 2^2) = 5 + 0.4 = 5.4 \text{ J}$ After $(\frac{1}{2} \times 0.4 \times 3^2) + (\frac{1}{2} \times 0.2 \times 6^2)$
 $= 1.8 + 3.6 = 5.4 \text{ J}$.

Since k.e = collision is elastic.

Elastic and inelastic collisions

When two objects collide momentum and energy are conserved. If some of the energy is converted into thermal energy, we say the collision is inelastic. In general, an inelastic collision is one in which there is a change of the internal energy of one or both of the colliding objects. If there is no change of the internal energy, the collision is elastic. All real macroscopic collisions are inelastic to some degree. However, in microscopic situations especially in elementary particles, elastic collisions are common because there is no change in the internal energy of such particles.

Examples.

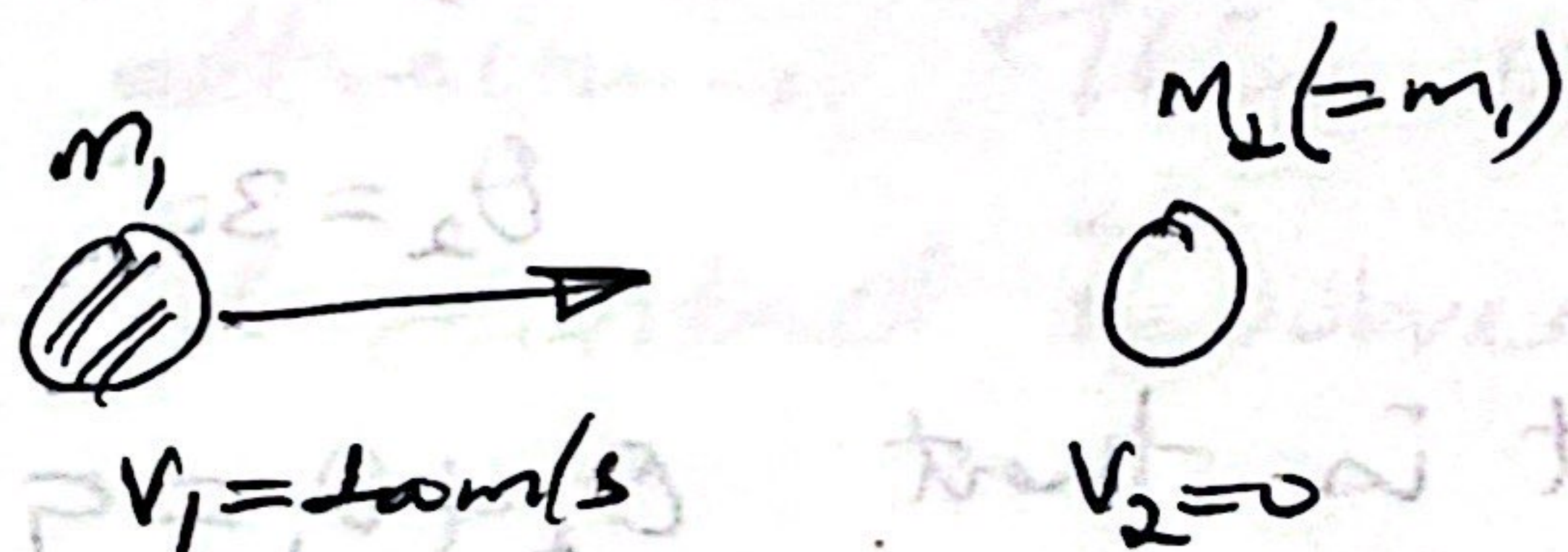


Example

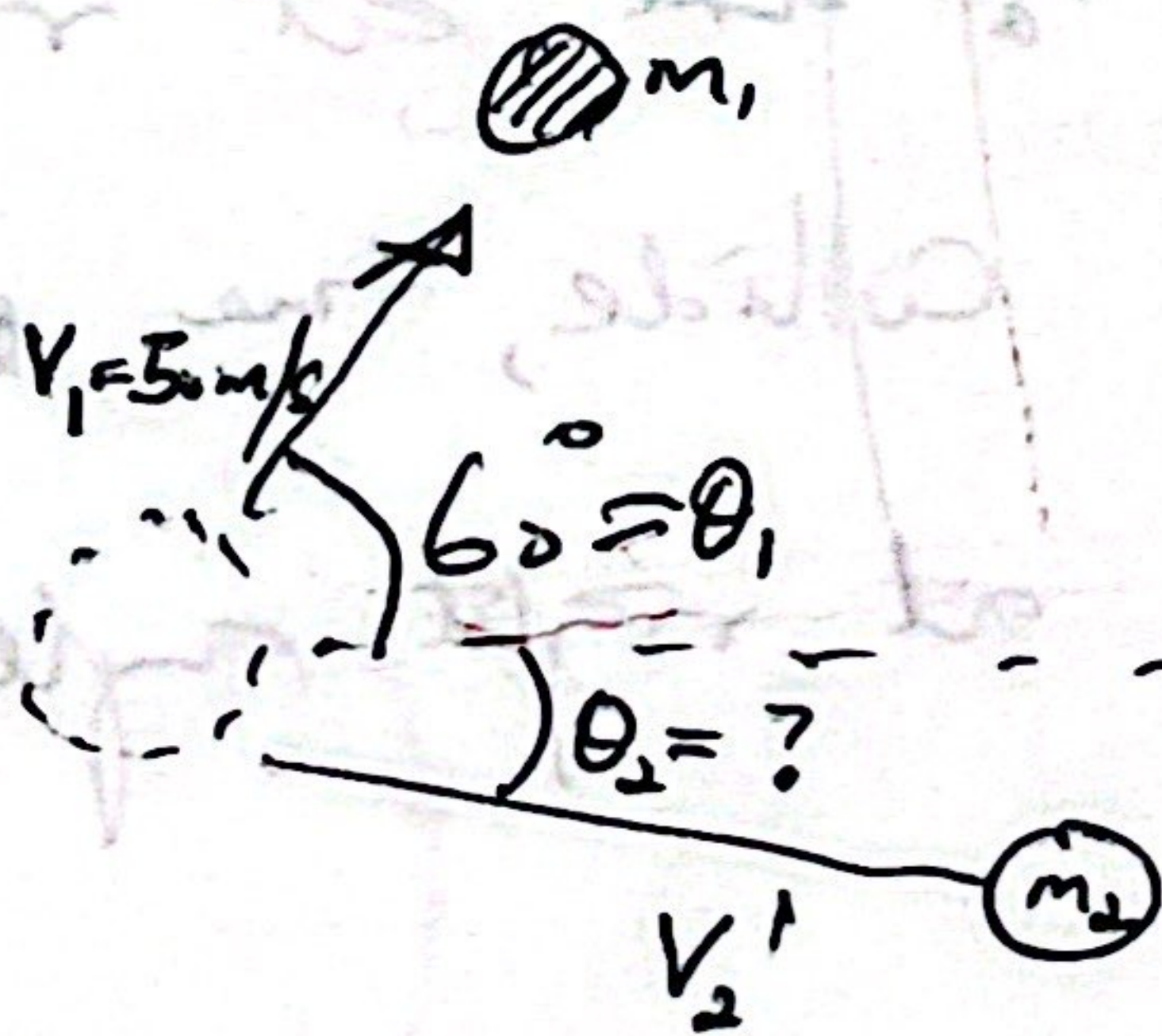
A ball is projected with a velocity of 10 m/s . The ball strikes another identical ball and is deflected through an angle of 60° . In what direction does the struck ball move and what is its velocity, assuming the collision to be elastic?

Sol.

Before



After



Before collision the x and y components of momentum are

$$p_x = m_1 v_1$$

$$p_y = 0.$$

After collision

$$m_1 v_1' \cos \theta_1 + m_2 v_2' \cos \theta_2 = m_1 v_1 \quad \text{X component}$$

$$m_1 v_1' \sin \theta_1 - m_2 v_2' \sin \theta_2 = 0 \quad \text{Y component}$$

Since the collision is elastic, we also have the energy equation

$$\frac{1}{2} m_1 v_1^2 \neq \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2$$

because $m_1 = m_2$ then

$$v_1' \cos \theta_1 + v_2' \cos \theta_2 = v_1$$

$$v_1' \sin \theta_1 - v_2' \sin \theta_2 = 0$$

$$v_1^2 = v_1'^2 + v_2'^2$$

After algebraic calculations

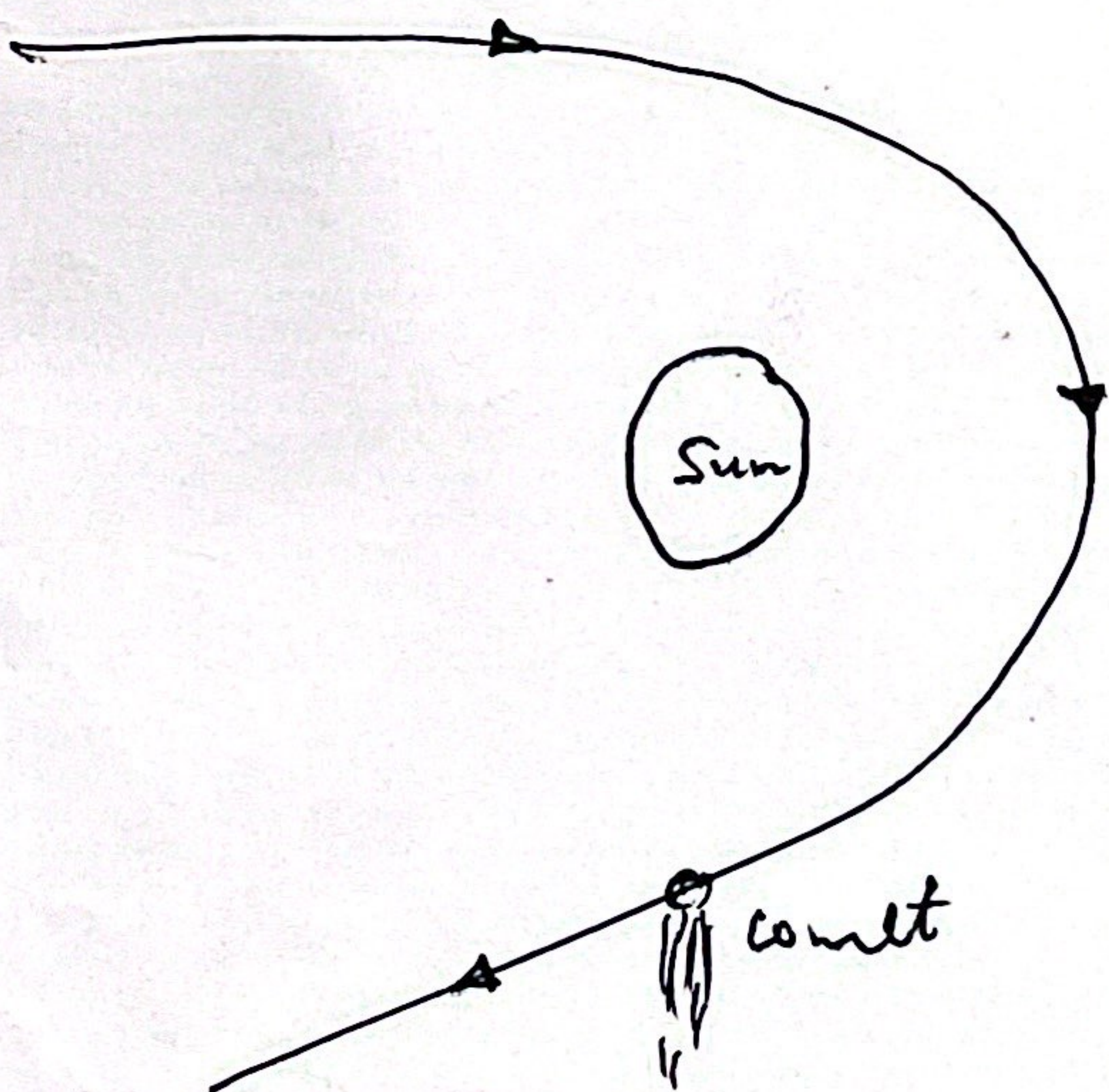
$$v_1' = 5 \text{ m/s}$$

$$v_2' = 8.66 \text{ m/s}$$

$$\theta_2 = 30^\circ$$

The important point is that $\theta_2 + \theta_1 = 90^\circ$, i.e. the two velocity vectors, v_1' and v_2' are at right angles. When two objects of equal mass collide, the objects always leave the point of collision at right angles. Eg. When two pool balls collide

* Collisions can be in two ways, Contact collisions and Contactless collisions. In contact collisions, the molecules of the colliding objects interact. The effect of a collision between two objects can be due to electrical, gravitational and nuclear forces that act between them. It is therefore possible to have a collision in which the objects never really come very close to one another. E.g. A comet is swept around the Sun by virtue of the mutual gravitational attractions. This is a form of collision with no physical contact involved.



Exercises

1. An object of mass 1 kg is moving in the x direction with $v = 45\text{ m/s}$ and another of mass 2 kg in y direction with $v = 15\text{ m/s}$. The two objects collide and stick together. Describe their motion.

Q 6 ^{Tu 11}
A ~~10 kg~~ Cart is moving over a horizontal surface at

- a constant $v = 2\text{ m/s}$. A 2 kg lump of clay is dropped into the cart from a height of 4 m and it sticks to the bed. Describe the subsequent motion. What ~~happened~~ happened to the vertical momentum? $[v = 1.67\text{ m/s}; \text{absorbed by the earth}]$

2. When a certain type of bullet ($m = 10\text{ g}$) is fired from a ~~rifle~~ rifle, a force of 3000 N is exerted for a millisecond. What is the muzzle velocity of the bullet? $[300\text{ m/s}]$

3. A fire hose ejects 50 kg of water/second at a speed of 40 m/s . What force must be exerted by the fireman holding the hose in order to keep it stationary?

□ In "shoot-'em-up" movies, the villain is often knocked down by a single bullet from the hero's gun. A .357 magnum bullet has a mass of approximately $10g$ and a muzzle velocity of $400m/s$. What would be the recoil velocity of an average size man struck by such a bullet? Is it likely that he would be knocked down?
 $\sim 6 \times 10^{-2} m/s$; Very likely

□ A head-on collision takes place between a $10000 kg$ truck travelling at $60 km/hr$ and a $1000 kg$ passenger car at $80 km/hr$, in the opposite direction. If the car and the truck lock together at the instant of collision, in what direction and at what speed is the wreckage travelling immediately after collision? What is the change of velocity of the passenger car? What is the change velocity of the truck?

Conservation of Linear momentum

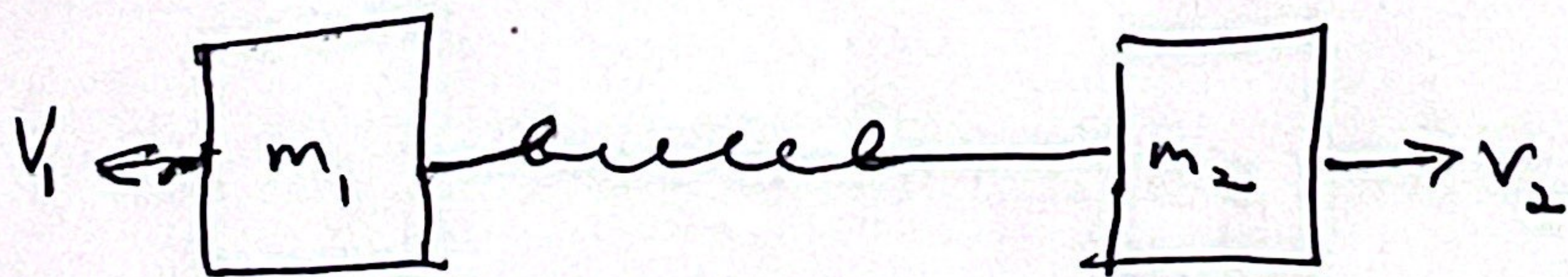
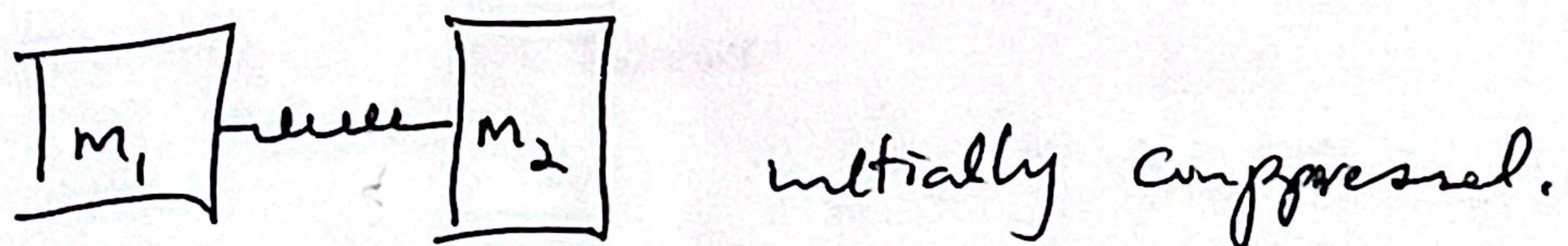
law — If there is no external force applied to a system, then the total linear momentum of that system remains constant in time.

This principle (or law) is valid both in classical or quantum mechanics. If a system consists of two objects, m_1 and m_2 , then

$$P_{\text{tot}} = P_1 + P_2$$

$$= m_1 v_1 + m_2 v_2 = \text{constant.}$$

Suppose



By the law

$$\begin{array}{ccc} \text{O} & = & P_1 + P_2 \\ \text{before} & & \text{after} \\ \text{release} & & \text{release} \end{array}$$

So that

$$m_1 v_1 = -m_2 v_2 \quad (\text{-ve sign indicate opposite } v_s)$$